DoangJoo "Alan" Synn

Ph.D. Student in Computer Science · Georgia Institute of Technology
Advised by Prof. HyunJoo Oh and Prof. Sehoon Ha
alansynn@gatech.edu · https://alansynn.com

Research Interests

My research builds high-efficiency and scalable machine learning systems, with a focus on automated dataloader tuning, memory-aware and distributed training, and privacy-preserving inference. I work across cloud and distributed computing to make ML training and deployment faster, more resource-efficient, and easier to use.

Education

Georgia Tech, Ph.D. in Computer Science *2022–Present*

- School of Interactive Computing; advised by Profs. Oh & Ha
- Research: computational design, mechanical artifacts, and motion synthesis (HCI/Graphics)

Korea University, M.Eng. in ECE *2020–2022*

- Thesis: *On Efficient Data Delivery for Machine Learning Systems*
- Advised by Prof. Jong-Kook Kim

Korea University, B.Eng. in EE *2012–2020*

- Alternative military service as a Software Engineer, 2016–2019

Publications

Refereed Conference Papers

1. **D. Synn** and J. Kim, "Num Worker Tuner: An Automated Spawn Parameter Tuner for Multi-Processing DataLoaders," *ACK*, Oct. 2021.
2. B. Ahn, **D. Synn**, M. Derkani, E. Ebrahimi, and H. Esmaeilzadeh, "Protopia AI: Taking on the Missing Link in AI Privacy and Data Protection," *NeurIPS*, 2021.
3. **D. Synn**, Z. Guo, S. Ha, and H. Oh, "MotionSmith: A Sketch-Based Design System for Automata Making," *CHI*, Apr. 2026.
4. J. Yu, P. Garg, **D. Synn**, and H. Oh, "Tangible-MakeCode: Bridging Physical Coding Blocks with a Web-Based Programming Interface for Collaborative and Extensible Learning," *CHI*, Apr. 2025.
5. J. Park, **D. Synn**, and J. Kim, "A Survey on the Advancement of Virtualization Technology," *KIPS*, May 2021.
6. D. Kim, **D. Synn**, and J. Kim, "SG-Drop: Faster Skip-Gram by Dropping Context Words," *KIPS*, Nov. 2020.

Refereed Journal Articles

1. M. Ryu, X. Piao, J. Park, **D. Synn**, and J. Kim, "ADaPT: An Automated Dataloader Parameter Tuning Framework using AVL Tree-based Search Algorithms," *TKIPS*, Jan. 2025.
2. **D. Synn**, X. Piao, J. Park, and J. Kim, "Distributed Training of Deep Learning Models Using Micro Batch Streaming," *KTSDE*, Mar. 2023.
3. X. Piao, **D. Synn**, J. Park, and J. Kim, "Enabling large batch size training for dnn models beyond the memory limit while maintaining performance," *IEEE Access*, 2023.

4 J. Han, **D. Synn**, T. Kim, H. Jung, and J. Kim, "Feature Based Sampling: A Fast and Robust Sampling Method for Tasks Using 3D Point Cloud," *IEEE Access*, Apr. 2022.

Preprints

1 J. Park, **D. Synn**, X. Piao, and J. Kim, "Dataloader Parameter Tuner: An Automated Dataloader Parameter Tuner for Deep Learning Models," *arXiv*, 2022.

Professional Experience

Ph.D. Research Intern, Dolby Laboratories *2024–2025*

- Built a time-series, multimodal datastore supporting low-latency media analytics in the Experience Delivery Lab.

MLOps Engineering Intern, LG Uplus Corp. *2023*

- Led **DeepCrunch**, a model-compression library supporting multi-backend AI inference for efficient edge deployment.

Research Scientist, Protopia AI Inc. *2021–2022*

- Built **Stained-Glass**, a privacy-preserving deep-learning framework enabling inference without raw-data exposure.
- Led framework development and Kubernetes/GPU infrastructure integration; the work contributed to a NeurIPS 2021 Demo.

Cloud Architect · AWS Educate Cloud Ambassador *2020–2022*

- Led engineering for four startup services, translating ambiguous product requirements into production backend systems.
- Built cloud-native infrastructure using Kubernetes, Terraform, CI/CD, observability, and cost-optimized autoscaling while mentoring early-career engineers.

Server Engineer, Flysher Inc. *2019*

- Shipped backend features for four social-game titles; one reached Facebook's global top 10.
- Built an ETL-to-Redshift analytics pipeline supporting data-informed feature rollout.

System Engineer, AKA Intelligence (Musio) *2017–2019*

- Designed embedded Linux boards and firmware for **Musio**, a deployed social robot.
- Coordinated hardware-software integration and production QA with overseas ODMs.

Teaching

Head TA, CS 4496/7496 Computer Animation *2024–Present*

- Georgia Tech, School of Interactive Computing.

Lecturer, Korea University *2021–2022*

- Taught introductory Java programming to 10+ undergraduate students.

Teaching Assistant, Korea University *2021*

- Digital System Design
- Data Structures and Algorithms

Lecturer, KAIST Engineering School *2012–2014*

- Co-developed national project-based coding education programs with South Korea's Ministry of Science and ICT.
- Designed and evaluated 20+ curricula combining Arduino, hardware prototyping, and software development for high-school and college students.

Invited Talks & Outreach

Invited Panelist and MotionSmith Demonstrator, AutomataCon *2026*

Motion, Making & Mechanical Design — GIFT Workshop, Georgia Tech CEISMC *2026*

Invited Talk, Korea University *2026*

MotionSmith Presentation, CHI *2026*

Honors & Awards

Fellowships

- Brain Korea 21 (BK21) Research Fellowship (2020–2022)

Awards

- Prize Winner, Creative Challenge Program @ Korea Univ. CTL (2015 and 2016)

Activities & Service

Georgia Tech Korean Student Association *2023–2024*

- President of Korean Students in the College of Computing.

Hardware & Software Community, Korea University *2014–2019*

- President; grew the community from 10 to 200+ active members across hardware and software tracks.

TEDxKoreaUniversity, Korea University *2015–2016*

- Founder & Chief Organizer; led a 47-person team in producing a 200-guest event with 7 speakers and a $4K budget.